IGCSE 0607 Practice Questions — E10.4 — Combined Questions & Markscheme

Averages and measures of spread

Total Questions: 6 — **Total Marks:** 42 Marks

Table of Contents & Quick Navigation

- **E10.4** Averages and measures of spread
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

---

1 A student records the number of points scored by a basketball team in 11 matches.

14 20 8 24 16 5 26 13 22 10 18

(a) Find the median.

..... **[1]**

(b) Find the range.

..... **[1]**

(c) Find the interquartile range.

..... **[2]**

(d) Calculate the mean.

..... **[2]**

2 The list shows the number of points scored by a basketball player in 7 games.

$$7 \quad 12 \quad 8 \quad 7 \quad 12 \quad x \quad 10$$

The mode of the number of points is 7.

(a) Write down the value of x.

$x =$ ..... **[1]**

(b) Find the range of the number of points.

..... **[1]**

(c) Find the median.

..... **[1]**

(d) Calculate the mean.

..... **[2]**

(e) In the 8th game, the player scores y points.
The new mean number of points for all 8 games is 9.5.
Find the value of y.

$y =$ ..... **[2]**

3 The table shows the heights, h cm, of 80 plants.

Height (h cm)	**Frequency**
$10 < h \leq 20$	12
$20 < h \leq 30$	25
$30 < h \leq 40$	28
$40 < h \leq 50$	11
$50 < h \leq 60$	4

(a) Write down the modal class.

..... **[1]**

(b) Calculate an estimate of the mean height.

..... cm **[4]**

4 The table shows information about the time, t seconds, taken by 100 students to complete a puzzle.

Time (t seconds)	Frequency
$10 < t \leq 20$	14
$20 < t \leq 30$	28
$30 < t \leq 40$	32
$40 < t \leq 50$	18
$50 < t \leq 60$	8

(a) Write down the modal time interval.

..... **[1]**

(b) Calculate an estimate of the mean time.

..... seconds **[4]**

(c) Calculate the percentage of students who took strictly more than 40 seconds to complete the puzzle.

..... % **[2]**

(d) Write down the interval that contains the upper quartile.

..... **[1]**

5 A factory produces 60 items.
The masses, m kg, of these items are recorded in the table.
The estimate of the mean mass is 48 kg.

Mass (m kg)	**Frequency**
$0 < m \leq 20$	p
$20 < m \leq 40$	14
$40 < m \leq 60$	16
$60 < m \leq 80$	q
$80 < m \leq 100$	5

(a) Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[5]**

(b) State the class interval which contains the median mass.
You must show your working.

..... **[2]**

6 The table shows information about the masses, m kg, of 50 parcels.

Mass (m kg)	Frequency
$0 < m \leq 2$	15
$2 < m \leq 4$	x
$4 < m \leq 6$	y
$6 < m \leq 8$	8
$8 < m \leq 10$	2

The estimate of the mean mass of the parcels is 3.8 kg.

(a) Use the total number of parcels to write down an equation in x and y, and show that it simplifies to $x + y = 25$.

[1]

(b) Use the estimate of the mean to show that $3x + 5y = 101$.

[4]

(c) Solve the simultaneous equations to find the value of x and the value of y.
You must show all your working.

$x =$ ..... , $y =$ ..... **[3]**

(d) Write down the interval that contains the median mass.

..... **[1]**

1 **(a)** 16 **[B1]**

(b) 21 **[B1]**

(c) Ordered list seen: $5, 8, 10, 13, 14, 16, 18, 20, 22, 24, 26$ **[M1]**

12 **[A1]**

(d) $(14 + 20 + 8 + 24 + 16 + 5 + 26 + 13 + 22 + 10 + 18) \div 11$ or $176 \div 11$ **[M1]**

16 **[A1]**

[Back to Question] *[Total Marks: 6]*

2 **(a)** 7 **[B1]**

(b) $12 - 7 = 5$ **[B1]**

(c) 8 **[B1]**

(d) $(7 + 7 + 7 + 8 + 10 + 12 + 12) \div 7$ or $\frac{63}{7}$ **[M1]**

9 **[A1]**

(e) $\frac{63 + y}{8} = 9.5$ or 9.5×8 **[M1]**

$76 - 63 = 13$ **[A1]**

[Back to Question] *[Total Marks: 7]*

3 **(a)** $30 < h \leq 40$ **[B1]**

(b) Uses midpoints $15, 25, 35, 45, 55$ **[M1]**

$\Sigma fx = (15 \times 12) + (25 \times 25) + (35 \times 28) + (45 \times 11) + (55 \times 4)$ or
$180 + 625 + 980 + 495 + 220 = 2500$ **[M1]**

$\frac{2500}{80}$ **[M1]**

31.25 **[A1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** $30 < t \leq 40$ **[B1]**

(b) Midpoints: $15, 25, 35, 45, 55$ **[M1]**

$\Sigma fx = 15(14) + 25(28) + 35(32) + 45(18) + 55(8)$ **[M1]**

$= 210 + 700 + 1120 + 810 + 440 = 3280$ **[M1]**

$\frac{3280}{100} = 32.8$ **[A1]**

(c) $18 + 8 = 26$ **[M1]**

26 **[A1]**

(d) $40 < t \leq 50$ **[B1]**

[Back to Question] *[Total Marks: 8]*

5 **(a)** $p + q = 25$ **[B1]**

$\frac{10p + 420 + 800 + 70q + 450}{60} = 48$ or $10p + 70q = 1210$ [B2] (Allow B1 for unsimplified $\Sigma fx = 10p + 420 + 800 + 70q + 450$ seen)

Correctly sets up and solves simultaneous equations to eliminate one variable **[M1]**

$p = 9$, $q = 16$ **[A1]**

(b) Identifies median is between 30th and 31st value and shows cumulative frequencies 23 and 39 (or $9 + 14 = 23$ and $23 + 16 = 39$) **[M1]**

$40 < m \leq 60$ **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** $15 + x + y + 8 + 2 = 50$ leading to $x + y = 25$ **[B1]**

(b) Midpoints: $1, 3, 5, 7, 9$ **[M1]**

$\frac{1(15) + 3(x) + 5(y) + 7(8) + 9(2)}{50} = 3.8$ **[M1]**

$15 + 3x + 5y + 56 + 18 = 190$ **[M1]**

$3x + 5y + 89 = 190$ leading to $3x + 5y = 101$ **[A1]**

(c) $3x + 3y = 75$ or substitution method $3(25 - y) + 5y = 101$ **[M1]**

$2y = 26 \implies y = 13$ **[A1]**

$x = 12$ **[A1]**

(d) $2 < m \leq 4$ **[B1]**

[Back to Question] *[Total Marks: 9]*